USB PD CTS ENGINEERING CHANGE NOTICE

NOTICE: Any Company or Companies submitting a USB Power Delivery ECR proposal must be one of the following: a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum. If a group of Companies is submitting an ECR proposal, each company must be either a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum.

SPECIFICATION REVISIONS AND ADDENDA: At any point in time, there shall only be one current version of the USB PD CTS, termed the production version. At the same time, there may also be proposed revisions to the specification's design which are not yet approved and shall be held confidential as deemed necessary by the USB 3.0 and USB 2.0 Promoters and within the Group of Working Committee(s).

PROCEDURES FOR SUBMITTING PROPOSALS: Both members of the USB Implementers Forum as a whole and members of the USB 3.0 and USB 2.0 Promoters may submit requests to revise the USB PD CTS Specification. Such a request may be rejected or may result in a USB PD Engineering Change Notice (ECN), which is the official way USB specifications may be changed.

FORMAT OF PROPOSAL: The originator of a request to alter the USB PD CTS Specification may do so by posting this to the USB Power Delivery Compliance working group for review. Once the proposal has been reviewed by the working group it will be passed to the USB 3.0 and 2.0 Promoters for approval to publish.

RESUBMISSION AND APPEAL: The originator of a request that was not approved can redraft the original request. Rewritten proposal will be treated as a new proposal and will be evaluated using the procedures described above. The originator of a request that was not approved can also submit an appeal to the USB 3.0 and 2.0 Promoters. The appeal must be made in writing and addressed to the Secretary of the USB Implementers Forum.

ABOUT THE ENGINEERING CHANGE REQUEST FORM:
The Purpose of this Engineering Change Request Form is to expedite the review process of the proposal by providing explanations, background information, and examples of the proposed changes at a high level. This form serves as an executive summary to the actual proposal.

STEPS ON HOW TO SUBMIT A USB PD ENGINEERING CHANGE REQUEST:

a. Please fill out the Engineering Change Request Form on the following pages completely:
    a. Detail the names and contact details for each of the ECR contributors
    b. Update the ECR Title
    c. Give a minimum of 2-3 sentences for each description on the form outlining the background to the ECR
b. For each section/table/figure to be updated:
    a. Detail the section number, starting page and figure/table number to be updated as appropriate.
    b. Detail existing text under "From Text"
    c. Detail changed text under "To Text"
c. Save the file as "USB PD CTS 1.0 R 1" followed by the ECR Title as per step a.b
d. Post the ECR in the USB PD CTS Documents section under "ECR | New ECRs".
    a. This ECR will then be reviewed by the Power Delivery Compliance Working Group.
    b. Revisions to the ECR originating from the review should be submitted as document revision of the original ECR using "Add new document".

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City: State/Province:

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USB PD CTS ENGINEERING CHANGE NOTICE

Title: TEST.PD.PS.SCR.1 :Multiple Request Load Test
Applied to: USB PD CTS Specification Q3 2026

Brief description of the functional changes proposed:
Ensure that source does not expect a keep alive while in SPR AVS contract

Benefits as a result of the proposed changes:
Source has been seen resetting after some time, corresponding to tPPSTiemout, due to bad implementation: issue shall be catch during compliance

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None if DUT properly implemented

An analysis of the hardware implications:
none

An analysis of the software implications:
none

An analysis of the compliance testing implications:
Tester SW shall be updated

An analysis of the Vendor Info File (VIF) implications:
None

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Actual Change Requested

(a). Section x.x.x, Page x, Figure/Table x-x

FROM:

Test.PD.PS.SRC.1: Multiple Request Load Test

Description: As a Sink only, the Tester verifies that the Source UUT responds correctly to various *Request* Messages.

Test Specific Tester Behavior: N/A

Test Conditions:

	Consumer Only, C/P	Provider Only, P/C, DRP
Rev2Src		✓
Rev3ChkdSrc		✓

Test Requirements: If Captive_Cable = YES, *vSrcNew* min, *vAvsNewmin*, or vPpsNew min limit shall be (*vSrcNew* or *vAvsNew* min) - (0.75*I/X) or (vPpsNew min) - (0.75*I/X), where I is the actual current being drawn, and X=3 if the sourced PDO contract is within the range (0, 3A], or X=5 if the sourced PDO contract is within the range (3A, 5A)

Test Procedures:

1. The Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.1.
2. The Tester repeats the following steps for each of the PDOs (specified in the VIF), starting with the first PDO:
    a. If the next Request will be for a PPS transition involving a current decrease (but no PDO change), the Tester reduces its current draw to the new level required, before sending the Request. (This avoids the current limit in the UUT shutting off V_{BUS}.) The Tester draws the requested Operating Current / Power with a current transition slew rate of 100mA per µs. The Tester monitors V_{BUS} voltage and the check fails if V_{BUS} voltage is not within the limits of *vPpsNew*.
    [TEST.PD.PS.SRC.1#9]
    NOTE: All measurements shall be within tolerances defined in the main specification and depicted in **Error! Reference source not found.**
    b. The Tester defines *Request* Message as following:
        i. B27 (Reserved) = 0b
        ii. B26 (Capability Mismatch) = 0b
        iii. B25 (USB Communication Capable) = 0b
        iv. B24 (No USB Suspend) = 1b
        v. B22 (EPR Capable) = 0b
        vi. For Fixed, Battery or Variable PDO:
            1. B19…10 (Operating Current / Power):
                a. For first Request: 0mA/0mW
                b. For subsequent Request: As defined in Step 2.f.i
            2. B9…0 (Maximum Operating Current / Power) = Maximum Current / Power in the Source PDO.
        vii. For PPS APDO:
            1. B20…9 (Output Voltage) = Maximum Voltage offered in Source APDO
            2. B6…0 (Operating Current):
                a. For first Request: 1A
                b. For subsequent Request: As defined in Step 2.f.ii
        viii. For SPR AVS APDO:

1. B20…9 (Output Voltage) – there are three iterations:
    a. 9V (minimum)
    b. 13.8V (an arbitrary value near the middle of the range)
    c. 20V if available, otherwise 15V (maximum)
2. B6…0 (Operating Current):
    a. For first Request: 0mA/0mW
    b. For subsequent Request: As defined in Step 2.f.i

The initial Request is for PDO #1 (5V Fixed) at 0mA. The complete list of Requests is defined in step 2.f.

TO:

Test.PD.PS.SRC.1: Multiple Request Load Test

Description: As a Sink only, the Tester verifies that the Source UUT responds correctly to various *Request* Messages.

Test Specific Tester Behavior: N/A

Test Conditions:

	Consumer Only, C/P	Provider Only, P/C, DRP
Rev2Src		✓
Rev3ChkdSrc		✓

Test Requirements: If Captive_Cable = YES, *vSrcNew* min, *vAvsNewmin*, or vPpsNew min limit shall be (*vSrcNew* or *vAvsNew* min) - (0.75*I/X) or (vPpsNew min) - (0.75*I/X), where I is the actual current being drawn, and X=3 if the sourced PDO contract is within the range (0, 3A], or X=5 if the sourced PDO contract is within the range (3A, 5A)

Test Procedures:

1. The Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.1.
2. The Tester repeats the following steps for each of the PDOs (specified in the VIF), starting with the first PDO:
    a. If the next Request will be for a PPS transition involving a current decrease (but no PDO change), the Tester reduces its current draw to the new level required, before sending the Request. (This avoids the current limit in the UUT shutting off V_{BUS}.) The Tester draws the requested Operating Current / Power with a current transition slew rate of 100mA per µs. The Tester monitors V_{BUS} voltage and the check fails if V_{BUS} voltage is not within the limits of *vPpsNew*. [TEST.PD.PS.SRC.1#9]
    NOTE: All measurements shall be within tolerances defined in the main specification and depicted in **Error! Reference source not found.**
    Tester shall wait for 500ms before issuing a new Request message unless other requirements apply
    b. The Tester defines *Request* Message as following:
        i. B27 (Reserved) = 0b
        ii. B26 (Capability Mismatch) = 0b
        iii. B25 (USB Communication Capable) = 0b
        iv. B24 (No USB Suspend) = 1b
        v. B22 (EPR Capable) = 0b
        vi. For Fixed, Battery or Variable PDO:
            1. B19…10 (Operating Current / Power):
                a. For first Request: 0mA/0mW
                b. For subsequent Request: As defined in Step 2.f.i
            2. B9…0 (Maximum Operating Current / Power) = Maximum Current / Power in the Source PDO.

vii. For PPS APDO:
    1. B20…9 (Output Voltage) = Maximum Voltage offered in Source APDO
    2. B6…0 (Operating Current):
        a. For first Request: 1A
        b. For subsequent Request: As defined in Step 2.f.ii

viii. For SPR AVS APDO:
To check that the DUT Source does not expect any keepalive functionality while the SPR AVS contract is active, Tester shall wait for tCtsAVSCheck after the PSReady message before issuing the next Request message for the next voltage.
    1. B20…9 (Output Voltage) – there are three iterations:
        a. 9V (minimum)
        b. 13.8V (an arbitrary value near the middle of the range)
        c. 20V if available, otherwise 15V (maximum)
    2. B6…0 (Operating Current):
        a. For first Request: 0mA/0mW
        b. For subsequent Request: As defined in Step 2.f.i

The initial Request is for PDO #1 (5V Fixed) at 0mA. The complete list of Requests is defined in step 2.f.

Appendix F: Timing table

To be added:

tCtsAVSCheck: min 20 sec